

MECH 191 — Metallurgy

Week 4

Carbon & Alloy Steels

September 14, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Steel Fundamentals

What steel is, why it matters, and how carbon controls everything

02

AISI/SAE Classification

How to read and use the four-digit steel designation code

03

Alloy Steels & Heat Treatment

Alloying elements, alloy steel families, CCT diagrams, and heat treatment processes

Module 3 · Week 4

Carbon & Alloy Steels

What to do this week:

Compare pearlite fraction across three carbon levels. Sketch pearlite lamellae from each specimen at 500×

Steps:

- 1 Observe — complete feature table
- 2 Sketch — label magnification & specimen
- 3 Answer — 4 short questions

Specimens to Check Out

- Specimen A — 1018 steel
 - Specimen B — 1045 steel
 - Specimen C — 1075 steel
-

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

Portfolio Handout — Module 3

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 ← here Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 4 — At a Glance

Topics

- Iron and steel — the difference carbon makes
- How carbon content controls strength, ductility, and weldability
- The carbon content spectrum: pure iron → low → medium → high carbon → cast iron
- AISI/SAE four-digit classification system
- Plain carbon steel grades and their applications
- Alloying elements (Mn, Cr, Ni, Mo, V, Si) and their effects
- When and why to specify an alloy steel over plain carbon

Resources

Reference Material

Moniz 5th — Ch. 11, 12 & 14
Iron & Steel / Carbon Steels / Heat Treatment

Kalpakjian 6th — Ch. 5 & 6
Metal Alloys & Ferrous Metals

Visualizer

Iron-Carbon Phase Diagram
(Interactive)

Reference

AISI/SAE Steel Designation Codes

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- 1 Why is low-carbon steel easy to weld while welding high-carbon steel requires preheat and post-weld heat treatment?
- 2 An AISI/SAE designation reads 4340. What does each part tell you about the composition and intended use?
- 3 Nickel, chromium, and molybdenum are all common alloying elements. What specific property does each primarily contribute?
- 4 Two steels with the same ultimate tensile strength: one plain carbon, one quenched-and-tempered alloy steel. Why might you choose the alloy steel?
- 5 You need steel for a cutting tool that must hold a sharp edge under high contact stress. What carbon content range and steel category should you specify?

Iron vs. Steel — What Carbon Does

Pure iron is a soft, weak metal with limited structural use. Steel is what happens when you dissolve carbon into iron — and control exactly how much.

Fe Pure Iron

- BCC crystal structure at room temperature (ferrite phase)
- Extremely soft — yield strength $\approx$ 130 MPa
- High ductility — almost too easy to deform
- No carbon = no lattice distortion = no dislocation blocking
- Rarely used structurally on its own

Iron is BCC below 912°C (ferrite) and FCC above (austenite). This allotropy is the foundation of all heat treatment — revisited in detail in Week 6.

Fe+C Steel

- Carbon dissolves interstitially into the iron lattice (like Week 1 alloys)
- Carbon atoms distort the lattice → block dislocation movement → stronger
- Yield strength ranges from 250 MPa to over 2,000 MPa
- Properties tunable through carbon content, alloying, and heat treatment
- 0.02% to 2.14% carbon by weight — a narrow window covering an enormous property range

Above 2.14% C, excess carbon can no longer stay in solution — it precipitates as iron carbide. This is cast iron, which behaves very differently and is covered in Week 5.

Carbon's Effect — The Central Tradeoff

Carbon is the most powerful and cost-effective strengthening tool in steel — but every increase is a trade-off. You cannot maximize all properties simultaneously.

Yield Strength	Hardness	Ductility	Toughness	Weldability
↑ increases	↑ increases	↓ decreases	↓ decreases	↓ decreases
More carbon → more lattice distortion → dislocations blocked more effectively → higher force needed to cause permanent deformation. Strength can be doubled or tripled with heat treatment.	Carbon increases resistance to surface indentation. Heat treatment (hardening) exploits this — high-carbon steels can be hardened to Rockwell C 60+, making excellent cutting edges and wear surfaces.	High carbon steels deform less before fracture. The distorted lattice resists dislocation movement — which gives strength but removes the plastic warning before failure. High-carbon steels snap, not bend.	Toughness requires both strength AND ductility. High carbon reduces the area under the stress-strain curve — less energy absorbed before fracture. High-carbon steels fail without absorbing significant impact.	High carbon creates hard, brittle heat-affected zones prone to hydrogen cracking. Carbon equivalent (CE) determines preheat requirements. Above ~0.40% CE, careful process control is essential.

Carbon Content — From Iron to Cast Iron

A narrow range of carbon content spans the entire world of iron-based materials. Each zone has its own properties, applications, and behavior.

Pure Iron	Low Carbon Steel	Medium Carbon Steel	High Carbon Steel	Cast Iron
< 0.02% C	0.02–0.30%	0.30–0.60%	0.60–2.14%	> 2.14% C
Applications:	Applications:	Applications:	Applications:	Applications:
Electrical cores, magnetic applications	Structural steel, sheet, pipe, fabrication	Shafts, gears, rails, automotive parts	Cutting tools, springs, wire rope	Engine blocks, brake rotors, pipe
Soft, ductile. Rarely structural.	Excellent weldability. Most produced steel.	Heat-treatable. Preheat to weld.	Hard, brittle. Difficult to weld.	Not steel. Covered in Week 5.

Steel Classification — AISI/SAE System

AISI and SAE jointly developed the four-digit designation system used throughout industry. Reading a code fluently is a core technician skill.

The Four-Digit Code:



Steel family &
primary alloying elements



Carbon content
in hundredths of %

Code	Family	Notes / Example Grades
10xx	Plain Carbon Steel	1020 — 0.20% C (low) / 1045 — 0.45% C (medium) / 1095 — 0.95% C (high)
41xx	Chromium-Molybdenum Steel	4140 — 0.40% C / excellent hardenability, the shop workhorse of alloy steels
43xx	Nickel-Cr-Molybdenum Steel	4340 — 0.40% C / very high strength, aerospace & defense, landing gear
31xx	Nickel-Chromium Steel	3160 — 0.60% C / improved toughness and hardenability
3xx	Austenitic Stainless (300s)	304, 316 — different naming convention; $\geq 10.5\%$ Cr — covered in Week 5

Reading the Code — Worked Examples

Being able to decode a steel designation is a core workplace skill — it tells you composition, expected properties, preheat needs, and heat treatment response instantly.

1020

Plain Carbon (10xx)
0.20% C

Low carbon steel. Very soft and ductile. Excellent weldability — no preheat needed for most thicknesses. Used for structural tube, sheet metal, pipe, and general fabrication. Cannot be significantly hardened by heat treatment.

1045

Plain Carbon (10xx)
0.45% C

Medium carbon steel. Good balance of strength and toughness. Can be quenched and tempered to improve strength. Preheat recommended for thicker sections. Common for shafts, bolts, gears, and tooling.

4140

Cr-Mo (41xx)
0.40% C

Chromium-molybdenum alloy steel. Same carbon as 1045 but far superior hardenability — can be hardened through thick sections. Excellent fatigue and wear resistance. Preheat required for welding. The workhorse of alloy steel.

4340

Ni-Cr-Mo (43xx)
0.40% C

Nickel-chromium-molybdenum alloy steel. Highest hardenability of common structural alloy steels. Used in aerospace, landing gear, and crankshafts. Very demanding preheat and post-weld heat treatment (PWHT) requirements.

Plain Carbon Steels — Grades and Applications

Plain carbon steels are the backbone of fabrication and manufacturing — simple, readily available, and well-understood. The right grade depends on the application.

Low Carbon Steel

< 0.30% C

- Soft and ductile — deforms without fracture
- Excellent weldability — no preheat required
- Good formability — bends, draws, and rolls easily
- Cannot be significantly hardened by heat treatment

Applications:

Structural sections (I-beams, angles, channels), sheet metal, pressure vessels, pipe, general fabrication, stampings

Typical: A36, A572, 1018, 1020

Medium Carbon Steel

0.30–0.60% C

- Stronger and harder than low carbon
- Can be quenched and tempered to improve strength
- Reduced weldability — preheat often needed
- Good balance of strength and toughness after heat treatment

Applications:

Shafts, gears, couplings, rails, automotive axles, bolts, crankshafts, agricultural equipment

Typical: 1040, 1045, 1050

High Carbon Steel

0.60–2.14% C

- Very high hardness and wear resistance
- Low ductility — limited formability
- Difficult to weld — requires careful preheat and PWHT
- Excellent response to hardening heat treatment

Applications:

Cutting tools, drill bits, springs, wire rope, saw blades, files, wear plates, knife blades

Typical: 1060, 1080, 1095

Alloying Elements — What Each One Does

Plain carbon steels cannot meet every requirement. Alloying elements are added to extend performance beyond what carbon alone can achieve.

Mn

Manganese

Improves hardenability and strength. Combines with sulfur to form MnS — trapping sulfur as harmless inclusions and preventing hot cracking during rolling.

Present in virtually all steels. Typical range: 0.5–2.0%.

Cr

Chromium

Increases hardenability, wear resistance, and corrosion resistance. At $\geq 10.5\%$ by weight, forms the passive chromium oxide layer that makes stainless steel corrosion resistant.

Key element in stainless steels and tool steels.

Ni

Nickel

Improves toughness, especially at low temperatures. Increases hardenability without reducing ductility significantly. Stabilizes the austenite phase (FCC) at room temperature.

Expensive. Used in cryogenic and austenitic stainless grades.

Mo

Molybdenum

Enhances hardenability and maintains strength at elevated temperatures. Reduces temper embrittlement. Improves pitting corrosion resistance alongside chromium (as in 316 stainless).

Key in 4140 alloy steel and 316 stainless. Often paired with Cr.

V

Vanadium

Refines grain size through precipitation hardening — fine vanadium carbide particles pin grain boundaries and resist grain growth. Improves both strength and toughness simultaneously.

Used in HSLA structural steels and high-speed tool steels.

Si

Silicon

Acts as a deoxidizer during steelmaking — scavenges oxygen from the melt. In higher quantities improves strength and electrical resistivity. Raises hardenability slightly.

Usually 0.1–0.3% in most steels as a steelmaking residual.

Alloy Steels — When Plain Carbon Is Not Enough

Plain carbon steels are the default — cheap, available, and well-understood. You specify an alloy steel only when the application demands something plain carbon cannot deliver.

Limitations of Plain Carbon Steel

- Cannot be hardened through thick sections — hardness drops off rapidly with depth (poor hardenability)
- Loses strength at elevated temperatures — not suitable for high-temperature service above ~300°C
- Corrodes readily in moisture, chemicals, and aggressive environments
- Poor toughness at low (sub-zero) temperatures — risk of brittle fracture
- Limited wear resistance for sliding or abrasion without surface treatment or coating

When to Specify Alloy Steel

- Deep hardening through thick sections — add Cr, Mo, or Ni to improve hardenability (4140, 4340)
- High-temperature service — Mo maintains strength above 400°C (Cr-Mo steels for pressure vessels)
- Corrosive environments — Cr $\geq 10.5\%$ for stainless; weathering steel for structural outdoor use
- Cryogenic or impact applications — Ni improves toughness at sub-zero temperatures
- Higher strength-to-weight ratio — HSLA grades with V or Nb for lighter structural designs

Steel in the Shop — What a Technician Must Know

Reading a material spec is only the start. Every steel grade carries specific implications for cutting, welding, heat treating, and inspection.

1

Identify the grade

Find the AISI/SAE code on the material certificate or drawing. First two digits: steel family and alloying elements. Last two: carbon in hundredths of a percent. Both matter for everything that follows — treat the spec as a full description, not just a name.

2

Assess weldability

Carbon equivalent: $CE = \%C + \%Mn/6 + (\%Cr + \%Mo + \%V)/5 + (\%Ni + \%Cu)/15$. CE above ~0.40% requires preheat. Above 0.60%, preheat is mandatory and PWHT is very likely required. The higher the CE, the more precisely heat input must be controlled.

3

Know heat treatment

Low carbon steels: no significant heat treatment response — use as-supplied. Medium and high carbon: quench to harden, temper to reduce brittleness. Alloy steels: same sequence but require more precise temperature control and should NOT be welded in the hardened condition without PWHT.

4

Inspect for the right defects

High carbon and alloy steels are prone to hydrogen (cold) cracking in the heat-affected zone — especially in thick sections or high-restraint joints. Understand WHY preheat and PWHT requirements exist in the procedure, not just that they do. That understanding is the difference between technician and craftsman.

CCT Diagram — What Actually Forms During Real Cooling

What the CCT Diagram Shows

- CCT = Continuous Cooling Transformation — plots temperature vs. time (log scale) with cooling curves overlaid
- Each steel composition has its own CCT diagram — the phase diagram is universal; the CCT is alloy-specific
- Cooling curves that intersect a transformation region produce that microstructure; faster curves bypass regions entirely
- Ms and Mf lines: martensite start and finish temperatures — steel must cool below Ms to begin forming martensite
- Hardenability: alloying elements (Cr, Ni, Mo, Mn) shift CCT curves right → martensite forms at slower cooling rates and in thicker sections

Microstructures by Cooling Rate

Pearlite (slow)

Furnace cooling / annealing. Ferrite + cementite lamellae. Soft, ductile, machinable. Used where formability and machinability matter.

Bainite (intermediate)

Interrupted quench or alloy steel air-cool. Finer than pearlite — better strength + toughness combination. Avoids martensite brittleness.

Martensite (fast)

Water or oil quench. BCC crystal — carbon trapped. Extremely hard (up to ~65 HRC) but brittle as-quenched. Must be tempered before use.

Heat Treatment Processes

Annealing

- Goal:** Soften, relieve stress, restore ductility
- How:** Heat to austenite region → furnace cool (very slow)
- Result:** Coarse pearlite — soft, ductile, excellent machinability
- Use:** After cold work, before machining, to restore formability

Normalizing

- Goal:** Refine grain size, homogenize microstructure
- How:** Heat to austenite region → still-air cool
- Result:** Fine pearlite — slightly harder and stronger than annealed
- Use:** Forgings and castings before further processing or machining

Quenching

- Goal:** Maximize hardness
- How:** Heat to austenite → rapid quench (water, oil, or polymer)
- Result:** Martensite — very hard (up to 65 HRC) but brittle
- Use:** First step in quench-and-temper; never used as-quenched alone

Tempering

- Goal:** Restore toughness after quenching
- How:** Reheat quenched martensite to 150–650°C → air cool
- Result:** Tempered martensite — lower hardness, far higher toughness
- Use:** After quenching — higher temp = more toughness, less hardness

Case Hardening

- Goal:** Hard surface over tough core
- How:** Carburize (add C to surface) or nitride (add N), then quench
- Result:** Hard martensitic case; tough ferrite/pearlite core
- Use:** Gears, cams, shafts needing wear resistance + impact toughness

Core Questions — Answers

1

Why is low-carbon steel easy to weld while high-carbon steel requires preheat and PWHT?

Higher carbon increases hardenability — the HAZ cools fast and forms brittle martensite. Without preheat/PWHT that's a fracture site. CE > 0.4 generally requires preheat.

2

An AISI/SAE designation reads 4340. What does each part tell you?

43 = Ni-Cr-Mo alloy steel family. 40 = 0.40% nominal carbon. Used in high-hardenability applications: aircraft landing gear, crankshafts, large structural components.

3

Nickel, chromium, and molybdenum are common alloying elements. What does each primarily contribute?

Chromium → hardenability & corrosion resistance. Nickel → toughness & low-temperature impact strength. Molybdenum → hardenability, creep resistance, prevents temper embrittlement.

4

Same UTS: plain carbon vs. quenched-and-tempered alloy steel. Why choose the alloy steel?

Alloy steel achieves that UTS at lower carbon → better weldability and toughness. Alloying elements give deep hardenability so full hardness develops through thick sections.

5

Steel for a cutting tool that must hold a sharp edge under high contact stress — what do you specify?

>0.6% C (typically 0.8–1.2%) for hard martensite. Tool steels add Cr, W, V for carbide particles that resist wear and retain hardness at elevated temperatures.

Key Takeaways

1

Steel is a family of alloys, not a single material

Carbon content is the single most important variable — controlling hardness, strength, ductility, and weldability. Every carbon increase trades strength for weldability and toughness.

2

The AISI/SAE code is a practical tool every technician must use

First two digits = steel family and alloying type. Last two = carbon in hundredths of %. Reading 4140 instantly communicates: Cr-Mo alloy steel, 0.40% carbon — good hardenability, preheat required to weld.

3

Alloying elements solve problems plain carbon steel cannot

Cr improves corrosion resistance and hardenability. Ni improves low-temperature toughness. Mo maintains high-temperature strength. Each solves a specific limitation — at the cost of price, weldability, and processing complexity.

4

Every grade carries welding and heat treatment implications

Carbon equivalent (CE) determines preheat requirements. Higher CE → higher preheat → more controlled heat input. Alloy steels may also require PWHT. Read the spec and know what to plan for before striking an arc.

5

Safety is not optional!

PPE is mandatory in the shop — no exceptions. We work with heat, sharp edges, and hazardous materials every lab session. Protect yourself and your classmates.

Resources & What's Next

Week 4 Resources

- Moniz Ch. 11, 12 & 14 — Iron, Carbon Steels & Heat Treatment
- Kalpakjian Ch. 5 & 6 — Heat Treatment & Ferrous Metals

Visualizers

- Iron-Carbon Phase Diagram (Interactive)
- CCT Diagram — Cooling Rate vs. Microstructure (Reference)

Coming Up — Week 5

Stainless Steels & Cast Irons

- What makes stainless steel corrosion-resistant — the passive chromium oxide layer
- The four stainless steel families and how microstructure defines each one
- Ferritic, martensitic, austenitic, and duplex — grades, properties, and selection
- Cast iron types — gray, white, ductile, and malleable — microstructure and applications